

MOVIES OTT PLATFORM

Technical Documentation

A full-stack OTT streaming platform with 30 Flyway migrations, 150+ REST endpoints, 5 user roles, and 3 frontend applications

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Backend: Java 21 + Spring Boot 3.4.5

Frontend: React 19 + Redux Toolkit + Tailwind CSS

Database: PostgreSQL 17 + Redis 7.4

Live API: <https://nmoviesapi.duckdns.org>

Frontend URL: <https://movies-app-complete.vercel.app>

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1. Executive Summary

MoviesApp is a comprehensive Internet streaming service (OTT) platform built on a modern architecture: a Java Spring Boot backend, React frontends, PostgreSQL for persistence, and Redis for caching. The platform provides distinct interfaces for super admins, admins, content managers, support staff, and end users, backed by full role-based access control (RBAC), JWT authentication with refresh-token rotation, TMDB integration, subscription management, and automated subscription-expiry notifications.

1.1 Key Metrics

Metric	Value
Java Files	214
JavaScript Files	59
Flyway Migrations	30 (V1-V30)
REST Endpoints	150+
User Roles	5 (User, Support, Content Manager, Admin, Super Admin)
Unit Tests	136 (JUnit 5 + Mockito)
Integration Tests	3 (TestContainers + PostgreSQL)
Email Templates	6 (Thymeleaf)
Frontend Pages	37 (15 user + 2 payment + 20 admin)
Redux Slices	4 (user, products, payment, notifications)
API Client Functions	59 (frontend)

2. Version History

This section tracks material changes to the platform and to this document across releases.

Version	Date	Summary of Changes
2.2.0	Jul 2026	Visual redesign for readability: accent-marked section headings, highlighted callout boxes for key notes, refreshed cover page; added missing resend-invite endpoint to API reference; exported alongside a PDF version.
2.1.0	Jul 2026	Documentation restructure: added version history, renumbered API reference into linked subsections, corrected currency symbol rendering, added table of contents.

Version	Date	Summary of Changes
2.0.0	Jun 2026	Added View History and Notification modules (migrations V29–V30); introduced token family + reuse detection for refresh tokens (V28).
1.5.0	Mar 2026	Added Watchlist feature and Content Manager role; multi-user refresh token support.
1.0.0	Nov 2025	Initial production release: core catalog, subscriptions, payments, admin panel.

3. System Architecture

The platform consists of three frontend applications, one backend API, and three infrastructure services.

3.1 Applications

Application	Framework	Port	Deployment
User & Admin Frontend	React 19 + Redux Toolkit + Tailwind CSS + MUI	3000	Vercel
Super Admin Frontend	React 18 + Vite + Tailwind CSS	3001	Vercel
Backend API	Java 21 + Spring Boot 3.4.5	8080	Docker on EC2

3.2 Infrastructure Services

Service	Image	Port	Purpose
PostgreSQL	postgres:17.6	5342	Primary database
Redis	redis:7.4	6379	Cache, JWT blacklisting, session store
Mailpit	axllent/mailpit	8025	Email testing UI (development only)

3.3 Backend Layer Breakdown

Layer	Count	Packages
Controllers	22	Controller/, Auth/
Services	27	Service/ (18 implementations + 9 interfaces)
Repositories	15	Repo/
Entities	20	Entity/
DTOs + Mappers	22	Dto/

Layer	Count	Packages
Records (DTOs)	22	Record/
Exceptions	23	Exception/ (21 custom + GlobalExceptionHandler)
Config	9	Config/ (seeders, RBAC, Swagger, scheduling)

| 3.4 CI/CD Pipeline (GitHub Actions)

Triggered on push to main when backend/ or docker-compose.yaml changes:

Step	Action
1. Build	JDK 21 setup, mvn clean verify (compile + 136 tests)
2. Docker	Build image, tag with commit SHA + latest, push to Docker Hub
3. Deploy	SSH into EC2, pull image, docker-compose down/up with --force-recreate
4. Cleanup	docker image prune -af + docker builder prune -af

4. Authentication & Security

| 4.1 JWT Authentication Flow

The system uses stateless JWT authentication with refresh-token rotation and device fingerprinting.

Step	Description
1. Login	User submits credentials via POST /api/v1/auth/login
2. Rate-limit check	In-memory ConcurrentHashMap tracks failed attempts per email
3. Authentication	Spring AuthenticationManager verifies credentials against the database
4. Principal resolution	CustomUserDetailsService searches Admin → Customer → ContentManager
5. Account validation	Checks isActive and isEmailVerified (Super Admin bypasses email check)
6. Token generation	Access token (15 min; Super Admin: 5 min) + refresh token (7 days)
7. Device binding	Refresh token bound to a SHA-256 hash of User-Agent + Accept-Language
8. Response	Returns accessToken, refreshToken, user DTO, and userType

| 4.2 Refresh Token Rotation with Theft Detection

Every token refresh performs security checks before issuing new tokens:

Check	Action if Failed
Device fingerprint mismatch	Theft detected → revoke all tokens for the user
Token already used (reuse)	Theft detected → revoke all tokens for the user
Token expired	Reject — user must re-login
Token not found	Reject — user must re-login

4.3 Rate Limiting

Threshold	Lockout Duration	Implementation
5 failed attempts	15 minutes	ConcurrentHashMap
15+ failed attempts	1 hour	Same ConcurrentHashMap, threshold check
Successful login	Resets counter	Removes entry from map

4.4 Additional Security Measures

- Token blacklisting: Redis-backed, keyed by JWT ID (jti), TTL = remaining token expiry
- OTP system: 6-digit code stored in Redis with a 5-minute TTL
- Password reset: UUID token stored in the database with a 3-hour expiry, emailed via a Thymeleaf template
- CORS: configured for Vercel, Render, and local development origins
- CSP headers: restrictive Content-Security-Policy with YouTube and ImgBB exceptions
- CSRF: disabled (stateless REST API)
- Session policy: STATELESS (no HTTP session)
- Password encoding: BCrypt

5. Role-Based Access Control

Five roles with hierarchical assignment policies and seven granular permissions.

5.1 Roles & Permissions

Level	Role	Permissions
1	ROLE_USER	MOVIE_READ
2	ROLE_SUPPORT	MOVIE_READ, USER_READ, VIEW_REPORTS
3	ROLE_CONTENT_MANAGER	MOVIE_READ, MOVIE_WRITE, MOVIE_DELETE

Level	Role	Permissions
4	ROLE_ADMIN	MOVIE_READ, MOVIE_WRITE, MOVIE_DELETE, USER_READ, USER_MANAGE, VIEW_REPORTS
5	ROLE_SUPER_ADMIN	All permissions (7 total)

5.2 Permission Details

Permission	Purpose
MOVIE_READ	View movies, shows, and TMDB content
MOVIE_WRITE	Create and update movies/shows
MOVIE_DELETE	Delete movies/shows
USER_READ	View user information
USER_MANAGE	Manage customers, admins, and roles
SYSTEM_CONFIG	Super Admin system configuration
VIEW_REPORTS	View analytics and reports

5.3 Role Hierarchy Policy

- canAssign(current, target): currentLevel > targetLevel (must be strictly higher)
- canModify(current, target): currentLevel >= targetLevel (equal or higher)

5.4 Endpoint Access Rules

Access Level	Endpoints
PUBLIC	/api/v1/ping, /api/v1/auth/login, /api/v1/auth/logout, /api/v1/auth/refresh-token, /api/v1/auth/customers, /api/v1/verify/email/**, /api/v1/validate/otp, /api/password-reset/**, /swagger-ui/**
PUBLIC (GET only)	GET /api/v1/tmdb/**, GET /api/v1/movies/**, GET /api/v1/shows/**
AUTHENTICATED	/api/v1/profile/**, /api/v1/auth/change-password, /api/v1/watchlist/**, /api/v1/notifications/**, /api/v1/view-history/**
MOVIE_WRITE	POST/PUT /api/v1/movies/**, POST/PUT /api/v1/shows/**
MOVIE_DELETE	DELETE /api/v1/movies/**, DELETE /api/v1/shows/**
USER_MANAGE	/api/v1/customers/**, /api/v1/roles/**, /api/v1/admins/**
SYSTEM_CONFIG	/api/v1/system/superadmin/**

6. Database Schema

PostgreSQL 17 with 30 Flyway migrations, 20 JPA entities, and comprehensive indexing.

6.1 Core Entities

Entity	Table	Key Fields	Relationships
Customer	customer	id, name, email, password, isSubscribed, imageUrl	ManyToMany(Role), OneToMany(Movie), OneToOne(UserPlanInfo)
Admin	admin	id, name, email, department, accessLevel, imageUrl	ManyToMany(Role)
ContentManager	content_manager	id, name, email, specialization, department	ManyToMany(Role)
Movie	movie	id, tmdbId, name, rating, poster, genre, category, ageRating	ManyToOne(Customer), ManyToOne(ContentManager)
Show	show	id, tmdbId, name, rating, poster, genre, category, ageRating	ManyToOne(Customer), ManyToOne(ContentManager)
Role	role	id, name (RoleName enum)	ManyToMany(Customer, Admin, ContentManager)
Payment	payment	id, transactionId, amount, paymentMethod, status	ManyToOne(Customer), ManyToOne(SubscriptionPlan)
UserPlanInfo	user_plan_info	id, isActive, startDate, endDate	ManyToOne(Customer), ManyToOne(SubscriptionPlan)
SubscriptionPlan	subscription_plans	id, planName, price, interval, description	—
RefreshToken	refresh_tokens	id, userId, userType, token, familyId, used	—
Notification	notification	id, title, message, category, messageType, read	ManyToOne(Customer)
WatchlistItem	watchlist_items	id, tmdbId, title, posterPath, mediaType	ManyToOne(Customer)
ViewHistory	view_history	id, tmdbId, mediaType, title, posterPath	ManyToOne(Customer, CASCADE)
PasswordResetToken	password_reset_token	id, token, email, expiry	—

6.2 Enums

Enum	Values
RoleName	ROLE_USER, ROLE_ADMIN, ROLE_SUPER_ADMIN, ROLE_CONTENT_MANAGER, ROLE_SUPPORT
Permission	MOVIE_READ, MOVIE_WRITE, MOVIE_DELETE, USER_READ, USER_MANAGE, SYSTEM_CONFIG, VIEW_REPORTS
UserType	CUSTOMER, ADMIN, CONTENT_MANAGER
IntentStatus	PENDING, COMPLETED, EXPIRED

6.3 Migration Summary (V1-V30)

Migrations	Purpose
V1-V8	Core tables: Customer, Movie, Show, Role, join tables, type columns
V9-V12	Subscription: plans, intents, payment, user_plan_info, schema alignment
V13-V16	Password reset tokens, refresh tokens, phone type change, show columns
V17-V21	Admin table, update triggers, cost removal, is_active flag, admin sequence
V22-V24	Categories, TMDB IDs, Content Manager table + join table
V25-V26	Multi-user refresh tokens, remove is_registered
V27-V28	Watchlist table, token family + reuse detection
V29-V30	Notification table, View History table

7. Features — User Side

7.1 Authentication

- Registration with Formik + Yup validation, avatar upload (ImgBB), and a password strength bar
- Email verification via a 6-digit OTP before account activation
- Login with role-based redirect (admin → /admin/dashboard, user → /)
- Forgot password with a time-limited token emailed via a Thymeleaf template
- JWT access token (15 min) + refresh token (7 days) with device fingerprinting

7.2 Content Discovery

- Home page: featured hero carousel + 8 horizontal content lists (Trending, Popular, Top Rated, Now Playing, Upcoming)

- Movies page: browse with search, sort (popularity/rating), and genre/year/rating filters via the TMDB discover API
- Shows page: same filter system as Movies, as a dedicated TV show catalog
- Detail page: backdrop + poster, rating, year, runtime, age rating, genres, cast, similar titles, trailer

| 7.3 User Actions

- Watchlist: add/remove/check with paginated views, filter/sort, and a play button
- Recently viewed: auto-tracked on detail-page visit, displayed as a horizontal row on the home page
- Share: copy link, WhatsApp, Twitter, and Facebook sharing from the detail page
- Notifications: in-app system with read/unread states, mark-all-read, and delete
- Profile: edit name/phone/address/avatar, change password with a strength meter
- Settings: notification toggle, dark mode, language selector (English/Hindi/Tamil/Telugu), delete account

| 7.4 Subscription & Payment

Plan	Price	Billing
Monthly	₹300	Billed every month
6-Month	₹50 / month	Billed every 6 months
Yearly	₹1,200	Billed annually

- Step 1: select plan → Step 2: email OTP verification → Step 3: payment checkout
- Payment form with Formik + Yup validation, card-number formatting, expiry, and CVV
- Success page with a Lottie animation, order details, and an auto-redirect countdown
- Automated subscription expiry: daily cron jobs deactivate expired plans and warn expiring plans (3-day notice)

| 7.5 Video Player

- Full-screen video player supporting YouTube embeds and direct video URLs with back navigation

8. Features — Admin & Super Admin

| 8.1 Admin Dashboard

- Welcome banner with user greeting
- Statistics cards: total users, movies, shows, admins
- User registration chart (Recharts) with monthly data
- Analytics dashboard with configurable chart views

- Quick-actions panel for common operations
- Platform metrics and system-status monitoring

8.2 Admin CRUD Operations

Entity	Create	Read	Update	Delete
Users	New user form	Searchable table + cards	Edit form + avatar	Delete with confirmation
Movies	Form with poster upload	Table + cards, stats	Edit with poster replace	Delete with confirmation
Shows	Form with all fields	Table + cards, stats	Edit form	Delete with confirmation
Admins	Invite via email	Table with role badges	Edit access level/status	Toggle active/inactive
Content Managers	Form + specialization	Table + analytics	Edit specialization	Delete with confirmation

8.3 Super Admin Panel

- Separate React 18 + Vite application with its own routing
- Dashboard: system overview, health check, admin/user counts, security status
- Invite Admin: 2-step form with validation, XSS prevention, email invitation
- Admins list: search, status toggle, CSV export, resend invite, detail modal
- Session management: 15-minute idle timeout, 8-hour absolute timeout, 2-minute warning modal
- Failed-login tracking: 5 max attempts with counter display
- Dark/light theme toggle

8.4 Subscription Expiry System

Job	Schedule	Action
checkExpiredSubscriptions	Daily, 9:00 AM	Deactivate expired subscriptions; send expiry email + in-app notification
checkExpiringSoonSubscriptions	Daily, 10:00 AM	Warn subscriptions expiring in 3 days; send warning email + notification

9. Email System

Dual email-provider architecture with profile-based switching.

9.1 Email Templates

Template	Trigger	Variables
otp-email	Registration / email verification	otp, email
password-reset-mail	Forgot password	resetLink
admin-invite-mail-simple	Super Admin invites an admin	toEmail, inviteLink
content-manager-invite-mail	Super Admin invites a content manager	toEmail, inviteLink
subscription-expiry-warning	3 days before expiry (automated)	email, planName, daysRemaining
subscription-expired	On expiry deactivation (automated)	email, planName

9.2 Provider Implementations

Provider	Profile	Transport	Use Case
SmtplibEmailService	dev	JavaMailSender (SMTP)	Local development via Mailpit
BrevoEmailService	prod	Brevo REST API	Production (SendinBlue)

10. API Reference

150+ REST endpoints grouped by domain. Full interactive documentation is available at </swagger-ui.html>.

10.1 Authentication (6 endpoints)

Method	Endpoint	Description
POST	/api/v1/auth/customers	Register new customer
POST	/api/v1/auth/login	Login (Customer/Admin/CM)
POST	/api/v1/auth/refresh-token	Refresh JWT token
POST	/api/v1/auth/logout	Logout (blacklist JWT)
POST	/api/v1/auth/change-password	Change password
POST	/api/v1/auth/set-password	Set password from invite

10.2 Movies & Shows (18 endpoints)

Method	Endpoint	Description
GET	/api/v1/movies	Get all movies
GET	/api/v1/movies/{id}	Get movie by ID
POST	/api/v1/movies	Create movie
PUT	/api/v1/movies/{id}	Update movie
DELETE	/api/v1/movies/{id}	Delete movie
GET	/api/v1/movies/search?q=	Search movies
GET	/api/v1/movies/category/{cat}	Movies by category
GET	/api/v1/movies/categories	All categories

The same 8 endpoints exist under /api/v1/shows, for 16 total, plus 2 additional show-specific endpoints (18 total).

10.3 TMDB Integration (14 endpoints)

Method	Endpoint	Description
GET	/api/v1/tmdb/search/movies	Search TMDB movies
GET	/api/v1/tmdb/search/shows	Search TMDB shows
GET	/api/v1/tmdb/trending/movies	Trending movies
GET	/api/v1/tmdb/trending/shows	Trending shows
GET	/api/v1/tmdb/top-rated/movies	Top rated movies
GET	/api/v1/tmdb/now-playing/movies	Now playing
GET	/api/v1/tmdb/upcoming/movies	Upcoming movies
GET	/api/v1/tmdb/movie/{tmdbld}	Movie details
GET	/api/v1/tmdb/tv/{tmdbld}	TV show details
GET	/api/v1/tmdb/movie/{tmdbld}/videos	Trailers
GET	/api/v1/tmdb/movie/{tmdbld}/cast	Cast
GET	/api/v1/tmdb/movie/{tmdbld}/similar	Similar movies
GET	/api/v1/tmdb/genres/movies	Movie genres
POST	/api/v1/tmdb/sync/movie/{tmdbld}	Sync from TMDB

10.4 Watchlist, Notifications & View History

Method	Endpoint	Description
POST	/api/v1/watchlist	Add to watchlist
GET	/api/v1/watchlist	Get watchlist
DELETE	/api/v1/watchlist/{tmdbId}/{mediaType}	Remove from watchlist
GET	/api/v1/watchlist/check/{tmdbId}/{mediaType}	Check if in watchlist
GET	/api/v1/notifications	Get notifications
GET	/api/v1/notifications/unread-count	Get unread count
PUT	/api/v1/notifications/{id}/read	Mark as read
PUT	/api/v1/notifications/read-all	Mark all as read
POST	/api/v1/view-history	Record a view
GET	/api/v1/view-history	Get recent views
DELETE	/api/v1/view-history	Clear history
DELETE	/api/v1/view-history/{tmdbId}/{mediaType}	Remove view

10.5 Subscriptions, Payments, Admin & Health

Method	Endpoint	Description
POST	/api/v1/subscription/intent/	Create payment intent
POST	/api/v1/payments/submitPayment	Process payment
POST	/api/v1/payments/subscribe-success	Mark as subscribed
GET	/api/v1/admins	Get all admins
POST	/api/v1/admins	Register admin
PUT	/api/v1/admins/{id}/toggle-status	Toggle admin status
GET	/api/v1/admin/stats/users	User registration stats
GET	/api/v1/admin/stats/content	Content stats
POST	/api/v1/system/superadmin/invite	Invite new admin
POST	/api/v1/system/superadmin/resend-invite	Resend admin invite
GET	/api/v1/ping	Simple health check
GET	/api/v1/health	Detailed health with DB status

11. Frontend Architecture

11.1 User & Admin Frontend (frontend/)

Technology	Version	Purpose
React	19.2.7	UI framework with modern hooks
Redux Toolkit	2.12.0	State management with Redux Persist
Tailwind CSS	3.4.19	Utility-first CSS framework
Material UI	9.2.0	Icon library and UI components
Axios	1.18.1	HTTP client with interceptors and auto-refresh
Formik + Yup	2.4.9 / 1.7.1	Form management and validation
Recharts	3.9.2	Chart library for admin dashboard
React Router DOM	7.18.1	Client-side routing
Swiper.js	14.0.5	Carousel/slider for featured content
Lottie React	1.2.10	Animation for success states

11.2 Page Count

Section	Count	Pages
User Pages	15	Home, Detail, Login, Register, ForgotPassword, Movies, Shows, Watchlist, Profile, Settings, Subscription, EmailVerification, VideoFullScreen, AboutUs, NotFound
Payment Pages	2	PaymentCheckout, Success
Admin Pages	20	Dashboard, AdminLogin, CMLogin, SetPassword, MovieList/Edit/New, ShowList/Edit/New, UserList/Edit/New, AdminList/Edit/New, CMList/Edit/New, Settings

11.3 Component Count

Category	Count	Key Components
Shared Components	16	NavBar, Footer, Sidebar, Featured, List, ListItem, FilterNavbar, WatchlistButton, ShareButton, RecentlyViewed, VideoPlayer, ErrorBoundary, GlobalLoader, NotificationDropdown
Admin Components	12	AdminLayout, AdminSidebar, AdminHeader, AdminProtectedRoute, Statistics, ChartSelector, AnalyticsDashboard, WidgetsLarge, PlatformMetrics, QuickActions, SystemStatus, AdvancedCharts

11.4 Redux State Management

Slice	State Key	Fields
userSlice	user	currentUser, authStatus, isFetching, error, errorMessage, successMessage
productsSlice	product	tmdbTrendingMovies, tmdbTrendingShows, tmdbFetching, error
paymentSlice	payment	userInfoAndSelectedPlan, paymentPlan, loading, error
notificationSlice	notification	notifications, unreadCount, loading, error

NOTE

*Slice file names in the codebase (userSlice.js, ProductsRedux.js, PaymentRedux.js, notificationRedux.js) are inconsistent; a naming-convention cleanup (e.g. standardizing on *Slice.js) is recommended for a future release.*

11.5 Design System

- Dark theme first: deep background (#050810) with glassmorphism effects
- Gradient accents: purple-to-cyan (#7c3aed → #06b6d4) for buttons, text, and decorative elements
- Ambient glows: floating blurred gradient circles as background decoration
- Scroll-reveal animations: IntersectionObserver-based with staggered delays
- Mobile-first responsive layout: separate mobile card layouts vs. desktop tables
- Custom scrollbar with a gradient thumb matching brand colors

12. Testing & Quality

12.1 Test Summary

Test Type	Count	Tools	Coverage
Unit Tests	136	JUnit 5, Mockito	12 services + 2 validation classes
Integration Tests	3	TestContainers + PostgreSQL	Payment, Movie, Customer flows
Test Data Factory	1	DataFaker	Realistic test data generation

12.2 Services with Unit Tests

Service	Test File	Test Count
MovieService	MovieServiceImplTest.java	12
ShowService	ShowServiceTest.java	13

Service	Test File	Test Count
CustomerService	CustomerServiceImplTest.java	24
AuthService	AuthServiceTest.java	5
PaymentService	PaymentServiceTest.java	3
PasswordResetService	PasswordResetServiceTest.java	10
RefreshTokenService	RefreshTokenServiceTest.java	14
RoleService	RoleServiceTest.java	11
SubscriptionService	SubscriptionServiceTest.java	8
SubscriptionPlanService	SubscriptionPlanServiceTest.java	4
PlanService	PlanServiceTest.java	2
CustomUserDetailsService	CustomUserDetailsServiceTest.java	4

12.3 Code Quality Tools

- Backend: Maven Spotless (formatting), Checkstyle (static analysis)
- Frontend: ESLint (linting), Prettier (formatting), webpack-bundle-analyzer
- All 136 tests pass with 0 failures, 0 errors, 0 skipped

13. DevOps & Deployment

13.1 Docker Compose Services

Service	Container	Image	Ports	Healthcheck
PostgreSQL	postgres-mv	postgres:17.6	5342:5432	pg_isready
Redis	redis-mv	redis:7.4	6379:6379	—
Mailpit	mailpit-mv	axllent/mailpit	1025 (SMTP), 8025 (UI)	—
Backend	movie-api	codenaren23/movies-api	8080:8080	depends on db + redis

13.2 Environments

Environment	Profile	Database	Email	Deployment
Development	dev	localhost:5342	Mailpit (SMTP)	Local Docker + Maven

Environment	Profile	Database	Email	Deployment
Production	prod	EC2 PostgreSQL	Brevo API	Docker on EC2

13.3 Deployment Architecture

- Backend: Docker container on AWS EC2, deployed via GitHub Actions CI/CD
- Frontend: Vercel (user/admin app + super admin), automatic deploys on push
- Database: PostgreSQL on EC2 with a persistent Docker volume
- Cache: Redis on EC2 (no persistence, ephemeral)
- DNS: DuckDNS for the backend API, a custom domain for the frontend
- SSL: Nginx reverse proxy with Let's Encrypt

14. Document Information & Support

Field	Detail
Maintainer	Narendran
Contact	naren06251999@gmail.com
Document Version	2.2.0
Last Updated	July 2026
Live API	https://nmoviesapi.duckdns.org
Live Frontend	https://movies-app-complete.vercel.app
API Docs (Swagger)	/swagger-ui.html